



Exam Yr.9 5.2 2019 - practice test

Maths standard (Ascham School)



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YEAR 9 5.2 MATHEMATICS
NOVEMBER EXAMINATION 2019

Reading time 5 minutes

Writing time 1.5 hours

Equipment Calculators may be used.

Total: $\frac{\quad}{115} = \quad \%$
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Instructions

- Write your **name** at the top of each page.
- Sections **A to F** are to be answered by all students. There are **18** numbered **pages**. Pages 16 to 18 have been left blank if extra working space is needed.
- Marks are awarded for **clear presentation** of solutions and **full working**.
- All answers should be completed in the **space provided** (including multiple choice)
- At the end of the exam, **hand in your exam as it is**.

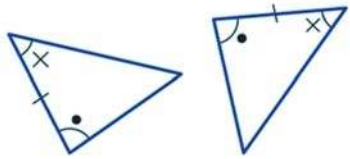
Exam Structure

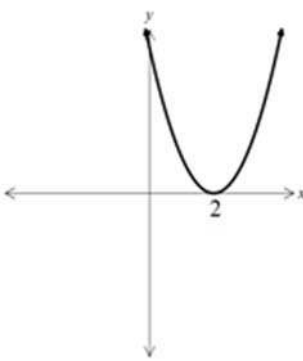
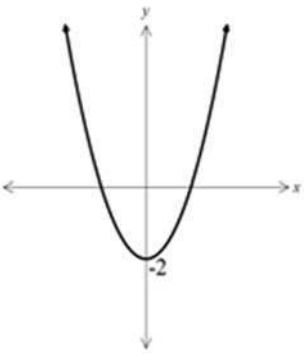
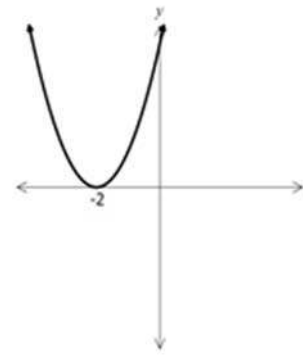
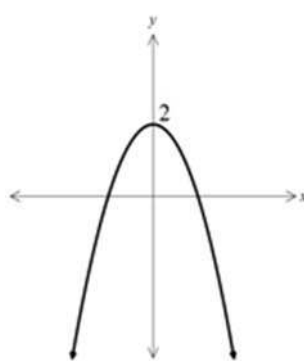
Section	Question Type	Marks
Part A	Multiple Choice	20
Part B	Short Answer	20
Part C	Free Response	28
Part D	Free Response	17
Part E	Free Response	20
Part F	Free Response	10
		Total = 115

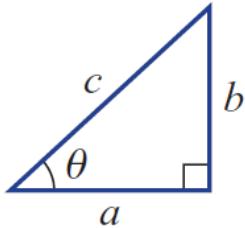
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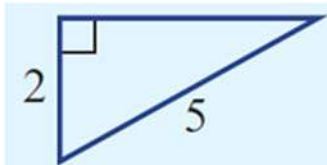
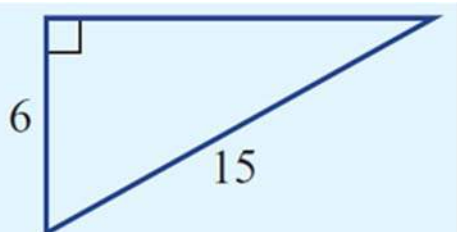
PART A Multiple Choice: Circle the letter of the correct response.
Each question is worth 1 mark.

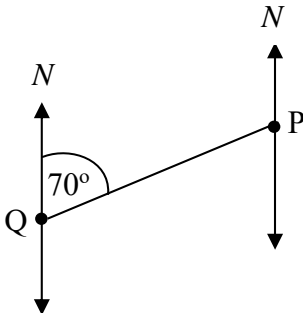
Total: <u>20</u>

1	Select the square root below that is also a surd. A $\sqrt{6}$ B $\sqrt{16}$ C $\sqrt{25}$ D $\sqrt{36}$
2	Find the median of these five numbers: 6 8 10 5 4 A 10 B 8 C 6 D 5
3	This season, Caitlin scored an average of 3 goals per match. If Caitlin played 12 matches over the season, how many goals did she score? A 3 B 4 C 12 D 36
4	Which of the following is a property of KITES? A Two pairs of adjacent sides are equal. B All angles are equal. C Diagonals bisect each other. D Opposite sides are parallel.
5	The triangles below are congruent. Which test proves it?  A SAS B RHS C SSS D AAS
6	A random sample of Year 9 Ascham students were asked to measure the circumference of their heads. Which type of data was collected? A Discrete B Continuous C Categorical D Census

7	$\frac{12a^2b^3}{4ab^2}$ is equivalent to: A $3ab$ B $8ab$ C $8ab^2$ D $4a^3b^5$
8	3^{-2} can be expressed as: A $\frac{1}{6}$ B $\frac{1}{3^{-2}}$ C $\frac{1}{9}$ D -6
9	250 000 in scientific notation is: A 25×10^4 B 2.5×10^5 C 2.5×10^6 D 2.5×10^{-6}
10	Which is the graph of $y = x^2 - 2$? A  B  C  D 

11	If the median of a set of scores is the 7th score, how many scores are there altogether? A 4 B 7 C 13 D 14
12	Snowflake the dog gained weight this year. Her mass increased by 10% and it is now 24 kg. What was Snowflake's mass last year? A 26.4 kg B 21.6 kg C 21.8 kg D 14 kg
13	Convert 36 km/h to metres per second: A 60 m/s B 6 m/s C 36 m/s D 10 m/s
14	Which of the following ratios is correct for the triangle below:  A $\tan \theta = \frac{b}{c}$ B $\tan \theta = \frac{a}{b}$ C $\tan \theta = \frac{b}{a}$ D $\tan \theta = \frac{c}{b}$
15	Sophie earns \$550 per week. Sally earns \$2200 per month. Which statement below is true? A Sophie earns more per year. B Sally earns more per year. C Both girls earn the same amount in one year. D It is not possible to determine who earns more.
16	Which of the following is a Pythagorean triad? A (1, 2, 3) C (1, 4, 9) B (9, 16, 25) D (3, 4, 5)

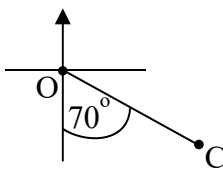
17	100 people were surveyed on the type of computers they used and the results are displayed below. What is the probability that a person selected at random owns a PC, but not a Mac?	<table><tr><td></td><td>Mac</td><td>No Mac</td><td>Total</td></tr><tr><td>PC</td><td>12</td><td>50</td><td>62</td></tr><tr><td>No PC</td><td>31</td><td>7</td><td>38</td></tr><tr><td>Total</td><td>43</td><td>57</td><td>100</td></tr></table>		Mac	No Mac	Total	PC	12	50	62	No PC	31	7	38	Total	43	57	100	A 12% B 50% C 31% D 62%
	Mac	No Mac	Total																
PC	12	50	62																
No PC	31	7	38																
Total	43	57	100																
18	What is the scale factor for this pair of similar figures?	Original  Image 	A $\frac{1}{3}$ B 3 C 9 D $\frac{1}{9}$																
19	Which of the following linear equations is INCREASING?	A $y = -x + 2$ B $y + x = 2$ C $y = 2 - x$ D $y - x = 2$																	
20	What is the bearing of Q from P?	A 290° B 250° C 200° D 070°																	



PART B Write your final answers in the columns provided.

Each question is worth 1 mark.

Total: 20

	Question and working	Answer only		Question and working	Answer only
1	Round correct to 3 significant figures: $3.1572727...$		11	Calculate the volume of a cube with side length 4 cm.	
2	Express in scientific notation: 0.005		12	Round $40^{\circ}12'40''$ correct to the nearest minute.	
3	Solve the equation: $x^2 = 25$		13	TRUE or FALSE?: “Opposite sides of a trapezium are equal”	
4	Simplify: $a^6 \times a^4$		14	Solve the equation: $\frac{24}{x} = 3$	
5	Simplify: $(y^2)^3$		15	Expand: $-2(4t - 5)$	
6	State the gradient of the line: $y = x - 5$		16	Expand: $(m + 2)(m - 4)$	
7	State the y-intercept of the line: $y = 4 - 3x$		17	Factorise: $9t - 81$	
8	A die is tossed. What is the probability of obtaining a number less than 3?		18	Simplify: $9p^0$	
9	What is the interior angle sum of a hexagon?		19	In a set of 7 scores, which score is the median?	
10	State the bearing of C from O. 		20	State the radius of the circle with equation: $x^2 + y^2 = 9$	

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PART C Answer in the space provided, showing all working.
Marks are indicated.

Total: <u>28</u>

1. Ava invests \$2000 for 4 years at an interest rate of 3% p.a.

a) How much does Ava have in her account at the end of the four years? (2 marks)
[Recall: $I = Prn$]

b) After how many years will Ava have obtained \$420 in interest? (2 marks)

2. Solve the equations:

i) $\frac{2x-1}{4} = 2$ (2 marks)

ii) $4(2m+3) = 32$ (2 marks)

3. The stem-and-leaf plot below shows finishing times (in minutes) for a group of 14 runners in the City-2-Surf race:

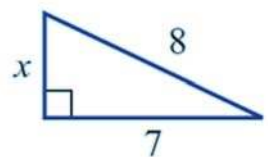
Stem	Leaf
5	3 5 8
6	1 4 7
7	0 2 4 7 9
8	2 6

a) Given the range is 33, find the missing score. (1 mark)

b) What is the median time? (1 mark)

c) What percentage of runners finished in **better** than 60 minutes? (2 marks)

4. Find the value of x as a surd. (3 marks)



PART C – Continued.

5. Izzy is paid \$950 per week.

- a) Find Izzy's annual salary.
[Recall: there are 52.18 weeks in a year for salary earners] (1 mark)

- b) Calculate Izzy's pay for 4 weeks. (1 mark)

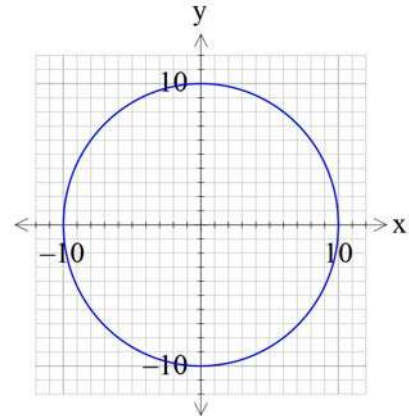
- c) Izzy is going to go on a 4 week holiday. As well as 4 weeks pay she is also entitled to an additional $17\frac{1}{2}\%$ holiday loading (on 4 weeks pay). Calculate her total holiday pay. (2 marks)

6. Factorise: $20xy - 10x^2$ (2 marks)

7. Expand and simplify: (2 marks)

$$(a - 7)(5 + a)$$

-
8. State the equation of the circle below:



(1 mark)

-
9. Consider the points $S(-3, 4)$ and $W(5, 0)$

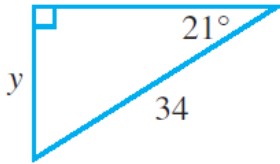
- a) Use the formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ to calculate the distance SW. Express your answer to 1 decimal place. (2 marks)

- b) Use the midpoint formula $MP: \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ to determine the midpoint of the interval SW. (2 marks)

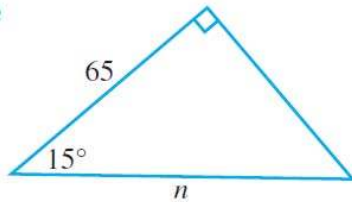
PART D Answer in the space provided, showing all working.
Marks are indicated.

Total: 17

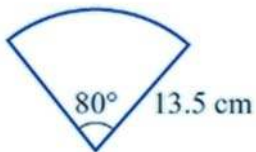
1. Find the value of the pronumeral to 1 decimal place. (2 marks)



2. Find the value of the pronumeral to 1 decimal place. (2 marks)



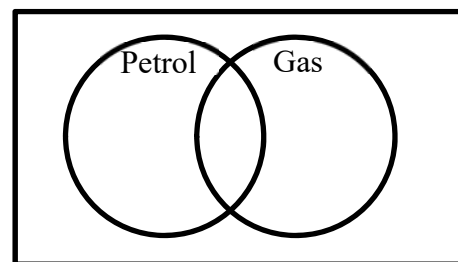
3. Calculate the perimeter of this sector correct to 2 decimal places. [Recall the **circumference of a circle** is given by the formula $C = 2\pi r$.] (3 marks)



4. 24 cars are tested for the type of fuel they run on.

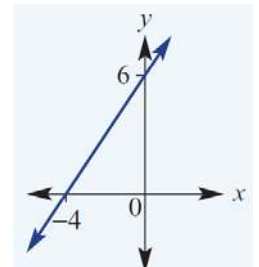
	Gas	Not gas	Total
Petrol	3	15	18
Not petrol	5	1	6
Total	8	16	24

- a) Copy this information into a Venn diagram: (2 marks)



5. Consider the line shown below:

- a) Calculate the gradient of the line. (1 mark)
[Recall $m = \frac{\text{rise}}{\text{run}}$]

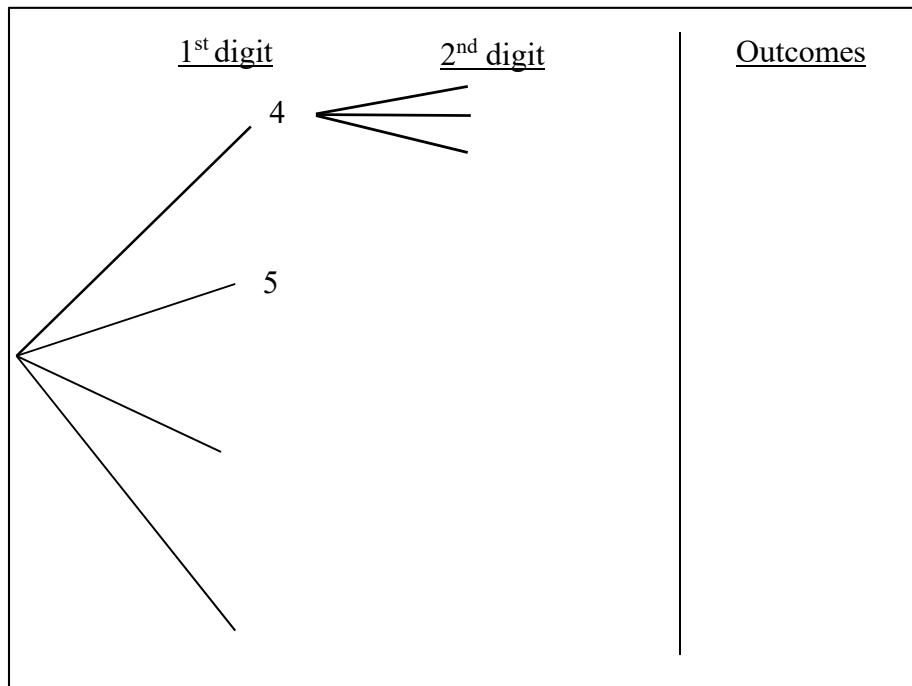


- b) Hence write the equation of this line in the form $y = mx + b$. (2 marks)

PART D – Continued.

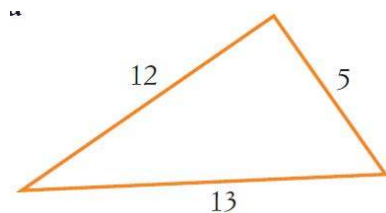
6. The numbers 4, 5, 6, and 7 are written on separate cards and placed in a bag. Two cards are drawn at random, one after the other **without replacement**, to form a two-digit number.

- a) Complete the tree diagram to show all possible two-digit numbers. (2 marks)



- b) What is the probability of drawing a number greater than 54? (1 mark)

-
7. Use the LHS/RHS method to prove that the triangle below is right angled. (2 marks)



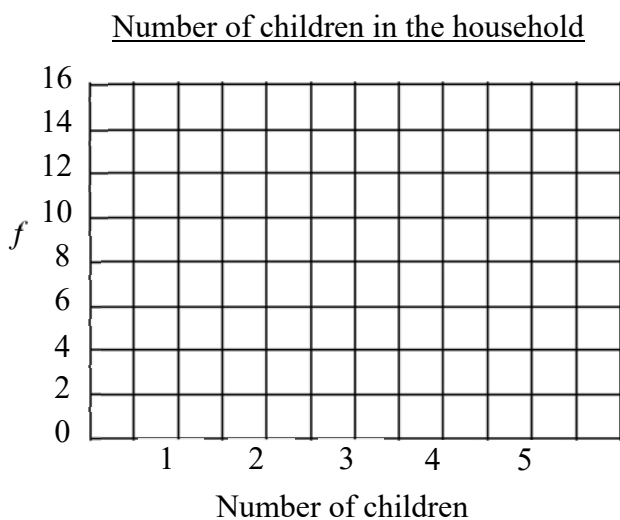
PART E Answer in the space provided, showing all working.
Marks are indicated.

Total: 20

1. A random sample of families were surveyed about how many children lived in the household.

No. of children	Frequency
1	5
2	15
3	7
4	3
5	1

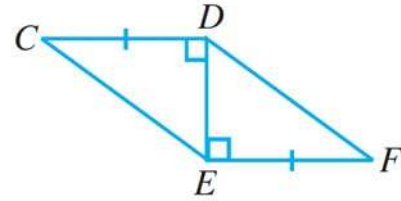
- a) Construct a frequency histogram below.
(2 marks)



- b) How many families were surveyed?
(1 mark)
- c) Is the distribution positively skewed, negatively skewed or symmetrical?
(1 mark)

2. Simplify: $\frac{h}{3} \div \left(\frac{5h}{2} \times \frac{4}{3h} \right)$ (3 marks)

3. Prove that $\triangle CDE \equiv \triangle FED$, giving reasons.
(2 marks)

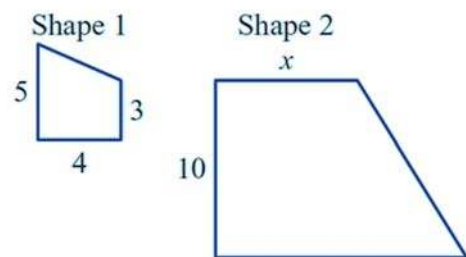


In $\triangle CDE$ and $\triangle FED$:

- _____
- _____
- _____

$\therefore \triangle CDE \equiv \triangle FED$ (_____)

4. The trapeziums below are known to be similar.



- a) Find the scale factor.
(1 mark)
- b) Hence find the value of the pronumeral.
(1 mark)

PART E – Continued.

5. The length of a rectangle is 7 cm longer than its width.
The perimeter of the rectangle is 94 cm.
Let w be the width of the rectangle. Write an equation and solve it to find w . (2 marks)
[Hint: a diagram is not necessary but it might be helpful].
-

6. Sophia's annual salary is \$129 800 and she also earns \$5432 in interest from a savings account. This year she makes allowable deductions of \$500 in charitable donations and \$1244 in work related expenses.

- a) Show that Sophia's taxable income is \$133 488. (1 mark)

- b) Using the tax table below, calculate Sophia's income tax payable. (3 marks)

Tax rates for Australian residents	
Taxable income	Tax on this income
0 – \$18 200	Nil
\$18 201 – \$37 000	19c for each \$1 over \$18 200
\$37 001 – \$80 000	\$3 572 plus 32.5c for each \$1 over \$37 000
\$80 001 – \$180 000	\$17 547 plus 37c for each \$1 over \$80 000
\$180 001 and over	\$54 547 plus 45c for each \$1 over \$180 000

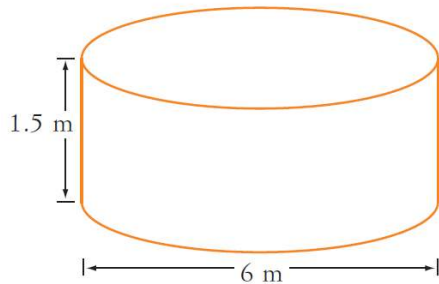
Source: © Australian Taxation Office for the Commonwealth of Australia

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7. Solve: $\frac{x}{7} + \frac{x-1}{4} = 0$ (3 marks)

PART F Answer in the space provided, showing all working.
Marks are indicated.

Total: <u>10</u>

1. A swimming pool is in the shape of a cylinder **open at the top**. The pool is 1.5 m deep and 6 m in diameter. The inside of the swimming pool is to be painted, including the floor.



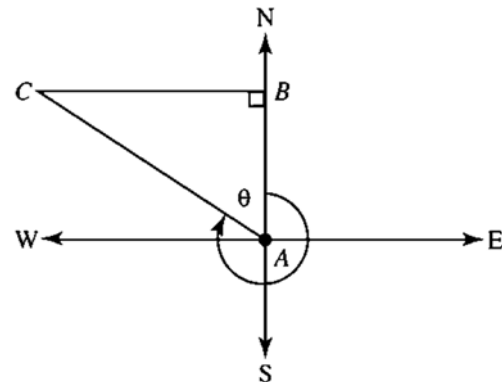
- a) Find the area to be painted, including the floor. Answer correct to 1 d.p. (2 marks)
[Recall the surface area of a **closed** cylinder is $A = 2\pi r^2 + 2\pi rh$]

- b) Find the volume of water required to fill the pool. Answer correct to 3 d.p.
[Recall the volume of a cylinder is $V = \pi r^2 h$]
(2 marks)

- c) How many litres of water are required to fill the pool? Answer correct to 3 significant figures. [Recall $1m^3 = 1kL$]
(2 marks)

2. Hattie drove due north from A to B for 50 km. She then turned west and drove to C, which is 72 km from B.

- a) Complete the diagram below with the information given. (1 mark)



- b) Find the size of the angle θ correct to the nearest degree. (2 marks)

- c) Hence find the bearing of C from A. (1 mark)

**** END OF EXAM ****

END OF EXAM – EXTRA WORKING SPACE
Clearly label the number of the question you are working on

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